Internal Representation of File

Inode

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Representing File…… Pre-requisite

Student should have knowledge of

- Kernel architecture of Unix operating system.
- Buffer cache structure & buffer cache maintenance algorithms.
- Basic file subsystem architecture & data structure.

Representing File …… Objectives

At the end of this chapter student will be able to understand

- Inode concept
- Types of Inode & Structure
- Allocation & release of incore inode
- Structure of a regular file.
- Conversion of byte offset to block number.
- Structure of a directory file.
- Conversion of pathname to Inode.
- Super block concept.
- Assignment & freeing disk inode.
- Allocation & release of disk block.

Representing File… Important Algorithms & processes

- Allocating & releasing of In-Core Inode. (iget & iput)
- Conversion of pathname to Inode. (namei)
- Allocating & releasing disk inode.(ialloc & ifree)
- Allocating & releasing disk blocks. (alloc & free)

Basic Understanding.....

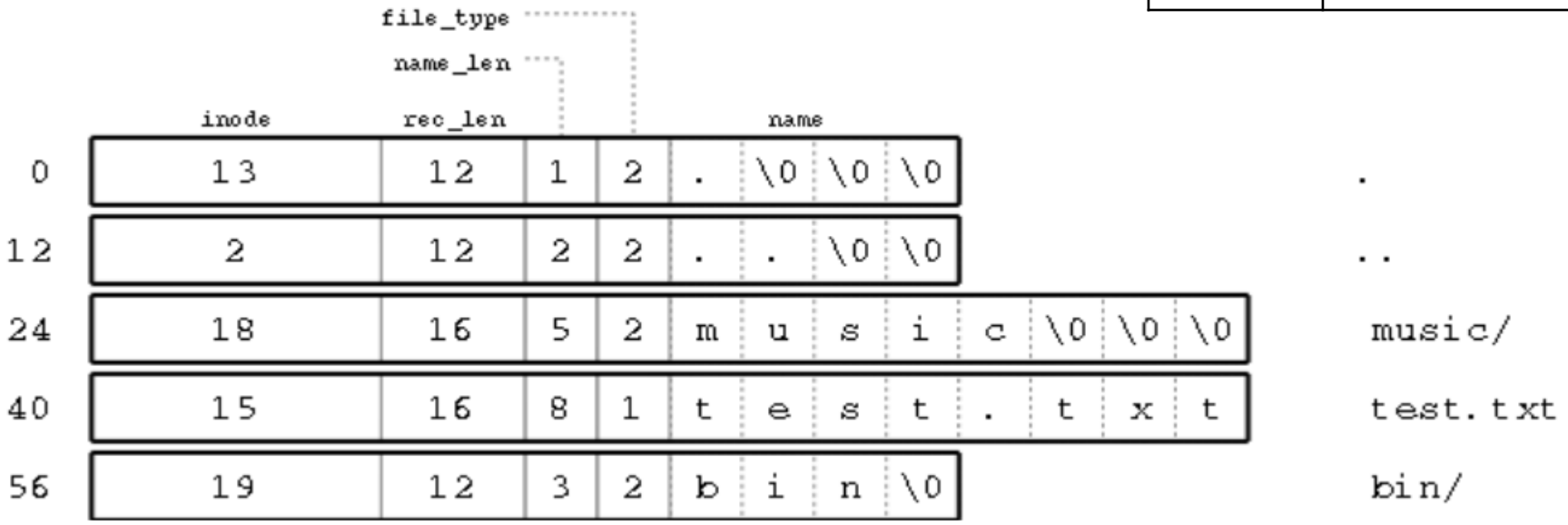
- Every file in a UNIX system has a *unique number*, called as --- *inode*.
- Processes interact with files using well defined system calls.
- The users specify a file with a character string which is the *file's path* and then the *system get the inode* which is mapped to the file that corresponds to the path.

Basic Understanding……

How files are stored???

```
struct dirent
{
        ino_t d_ino; /* inode number */
        off_t d_off; /* offset to the next dirent */
        unsigned short d_reclen; /* length of this record */
        unsigned char d_type; /* type of file
        char d_name[256]; /* filename */
};
```

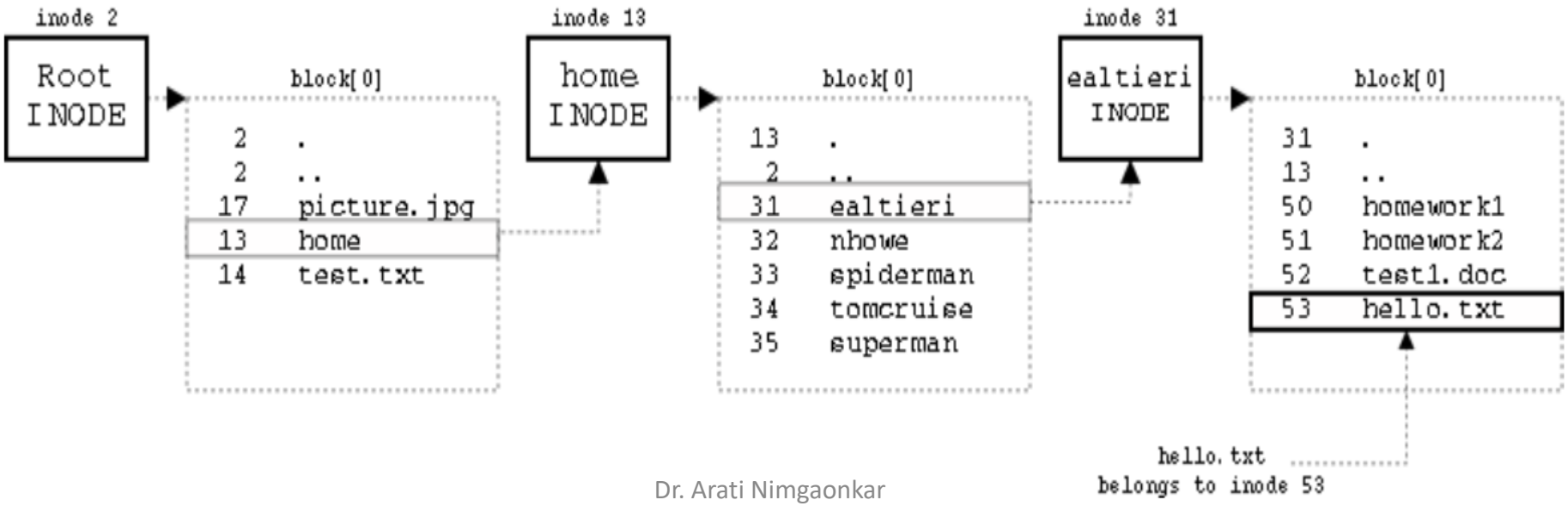
file_type	Description
0	Unknown
1	Regular File
2	Directory
3	Character Device
4	Block Device
5	Named pipe
6	Socket
7	Symbolic Link



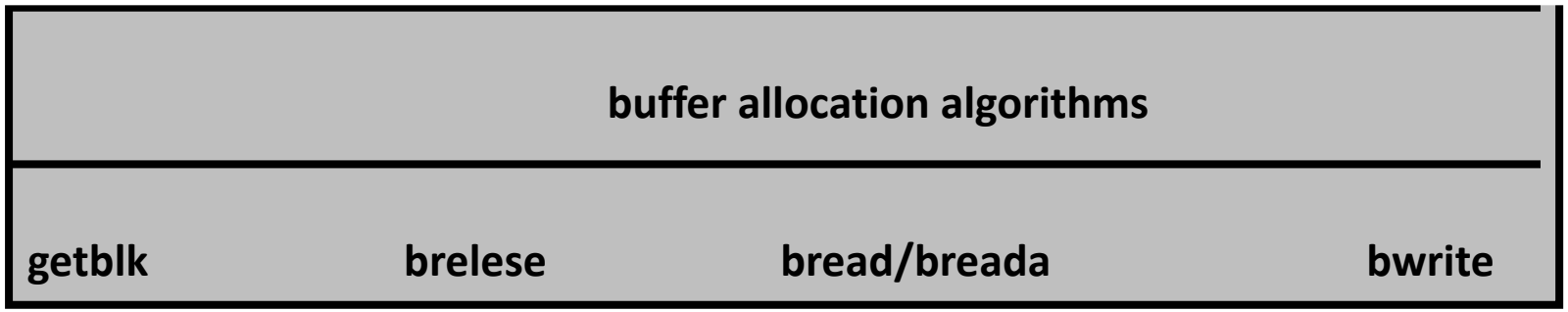
Basic Understanding……

Filename	Inode
.	13
..	2
music	18
test.txt	15
bin	19

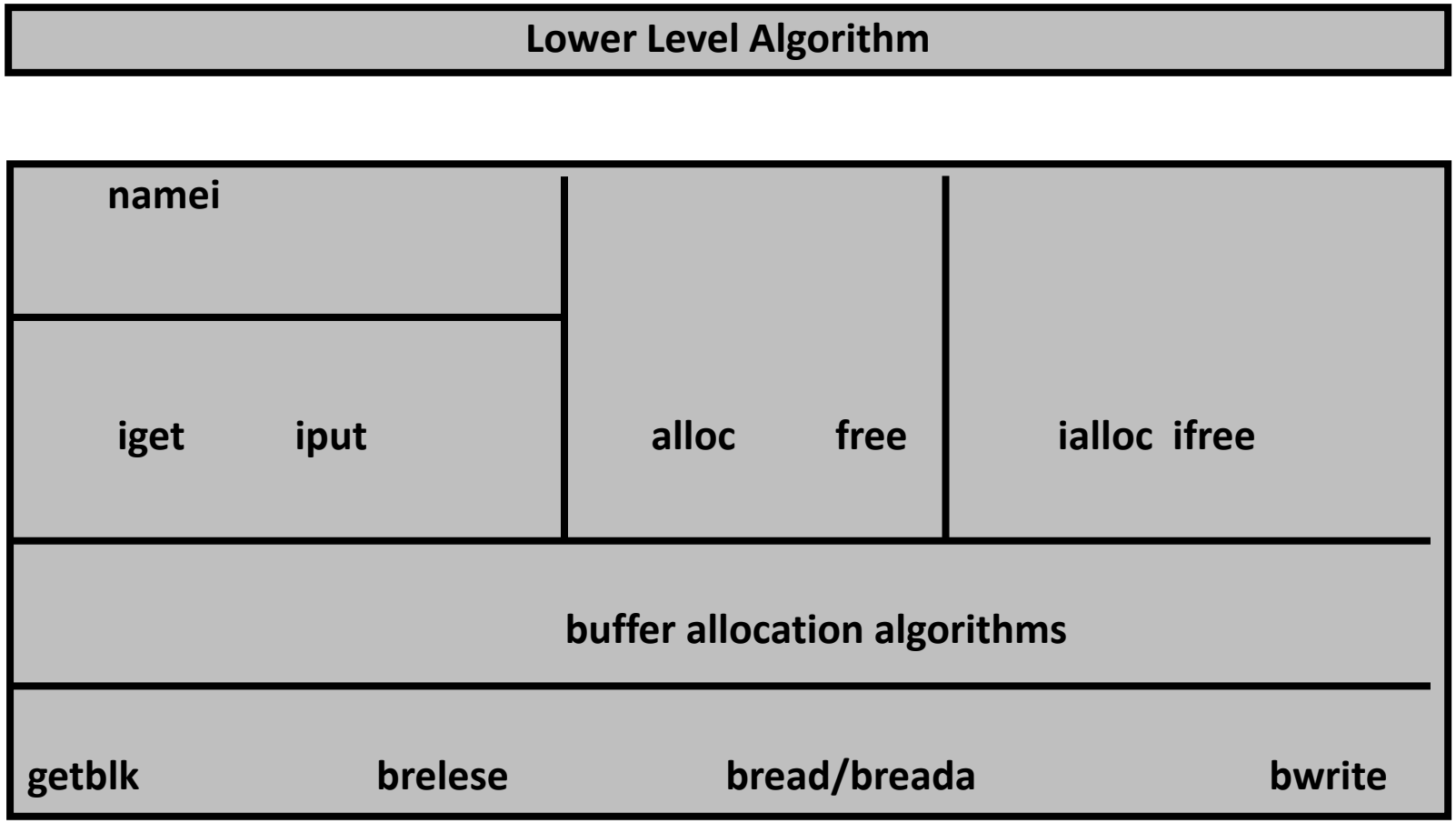
```
test.txt
    |
    | directory lookup
    v
inode 15
    |
    +---- permissions
    +---- owner
    +---- size
    +---- timestamps
    +---- data block addresses
```



File subsystem… overview

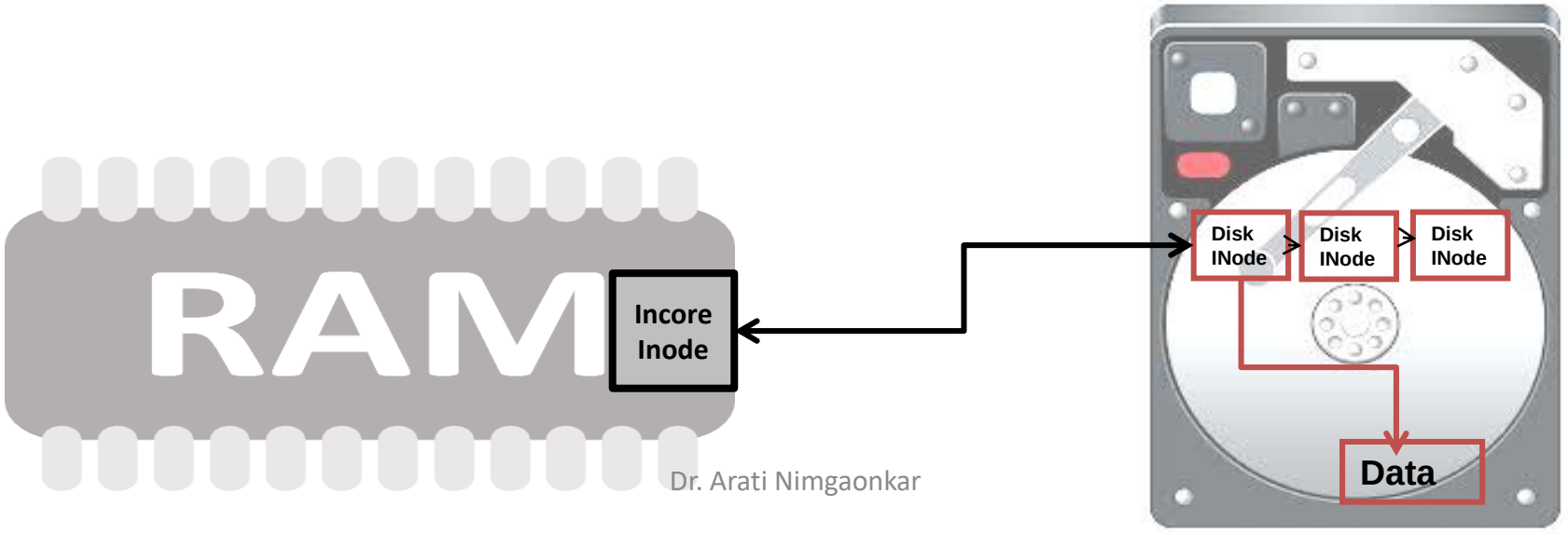


File subsystem… overview



Representing File… Inode

- Number assigned to each file on the machine.
- Internally kernel refer each file as a inode.
- Index Node or Information Node
- Two types of the Inodes
  - Disk Inode – Stores static information.
  - In-core Inode – Stores dynamic information.

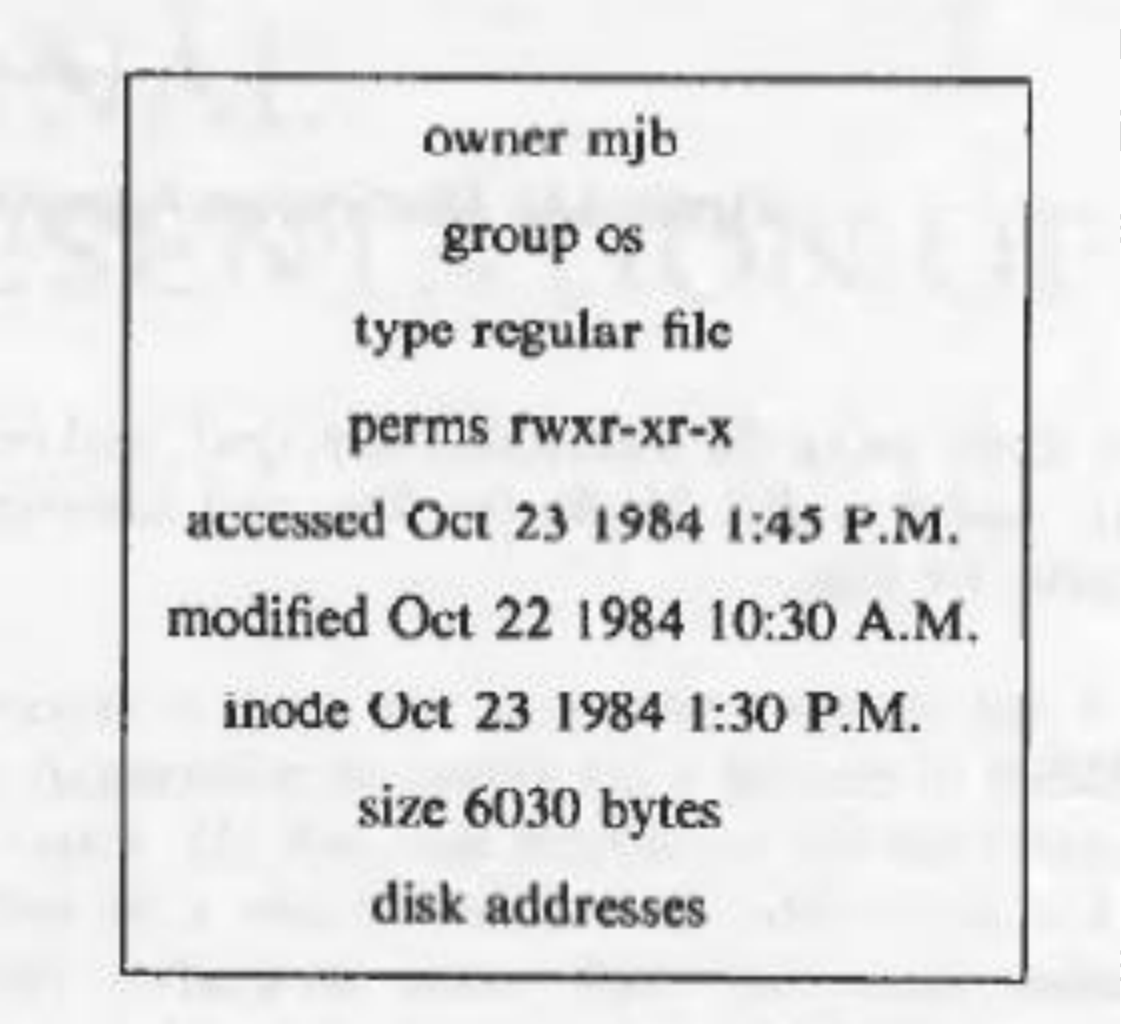


Representing File… Disk Inode - Fields

1. **File owner**….. can be individual owner or group owner.
2. **File Type**…… can be regular, directory, device file, pipe
3. **File access permissions**….. can be combination of read, write, execute.
4. **File access times**…… can be
    - last modification time of a file
    - last accessed time of a file,
    - Time when inode was last modified .
5. **Number of links to file**. – number of names the file has.
6. **Table of contents** of disk addresses where actual data is stored.
7. **File size** .

Representing File… Disk Inode - Fields

1. File own… …ner.
2. File Typ… …ipe
3. File acc… …ead, write, execute
4. File acc…
   - …
   - …
   - …
5. Numbe… … has in that director…
6. Table of… …ata is stored.
7. File size .



Representing File… Disk Inode



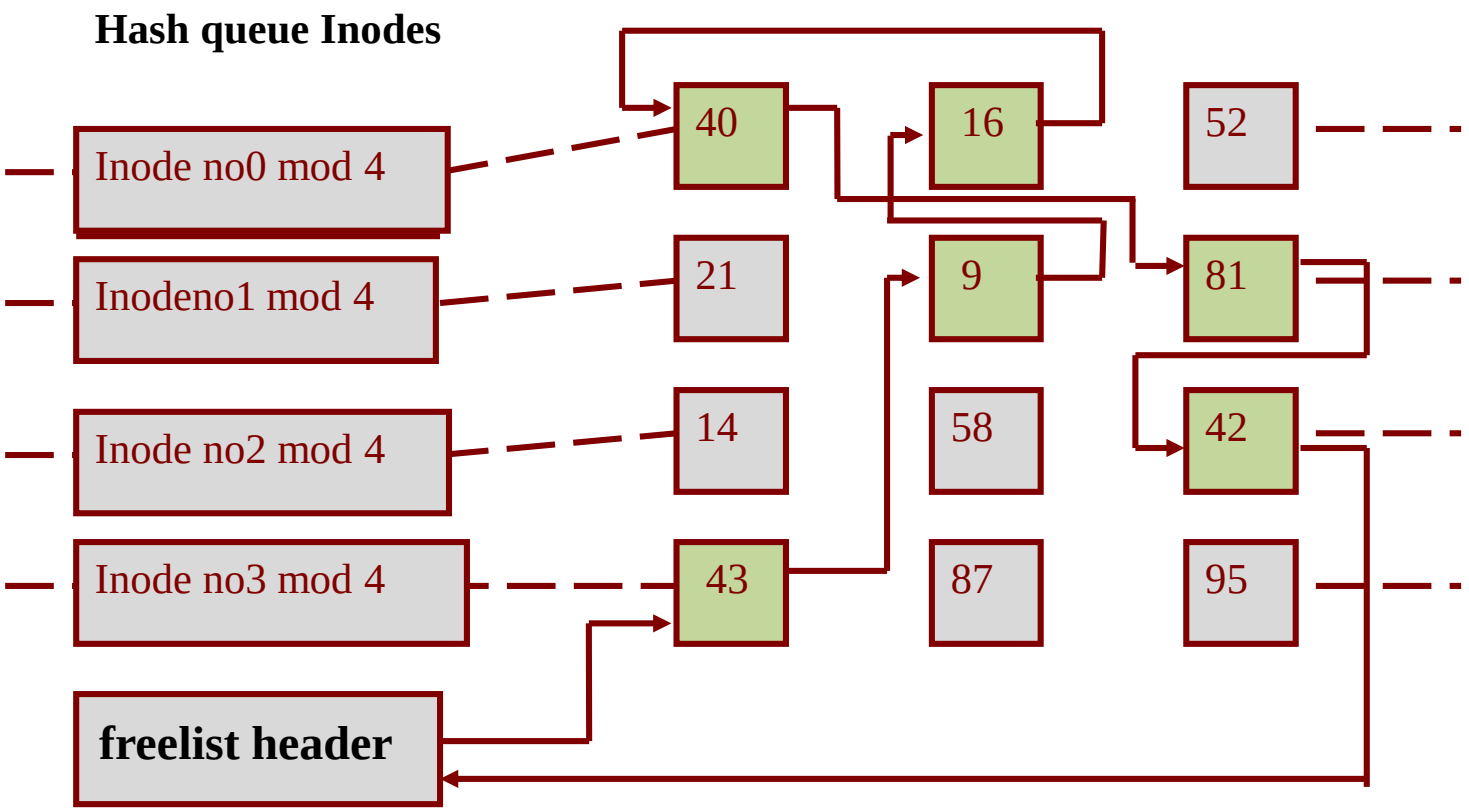
Changing the inode does not imply that the contents of the file are changed……???

Representing File… In-core Inode

In-core Inode contains following fields in addition to the disk inode.

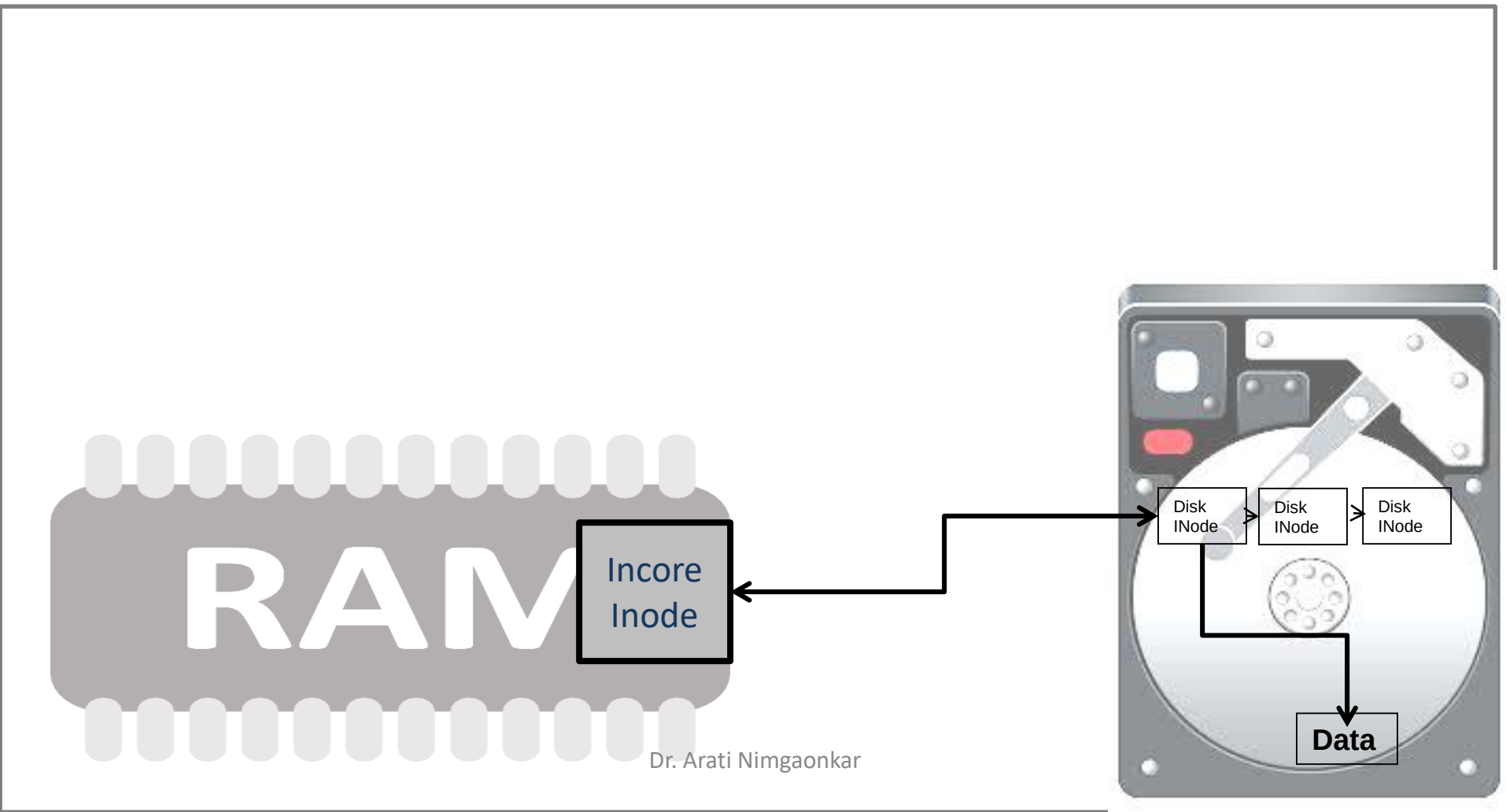
8. **Status** - can be inode is locked.
   - process is waiting for an inode which is locked.
   - in-core inode differs from disk inode as result in change in data
   -in-core inode differs from disk inode as result in change in file data
   - file is a mount point.
9. **Device no of the file system containing that file**
10. **Inode number.**
11. **Reference count-**Indicating no. of instances of the file those are active
12. **Pointer to other in-core inodes -** kernel links inode on hash queue & on a free list in same way as that of buffers.

Representing File… In-core Inode structure



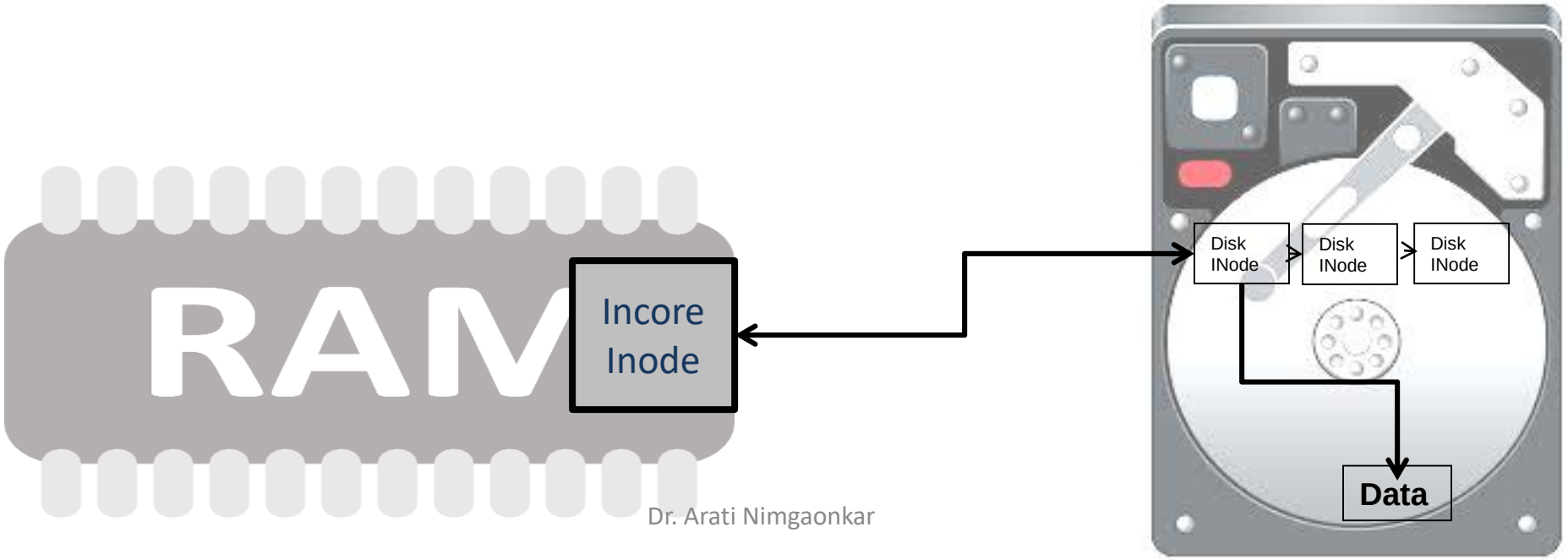
Algorithms related to inode

- Allocation of in-core inode……iget
- Release of in-core inode……iput



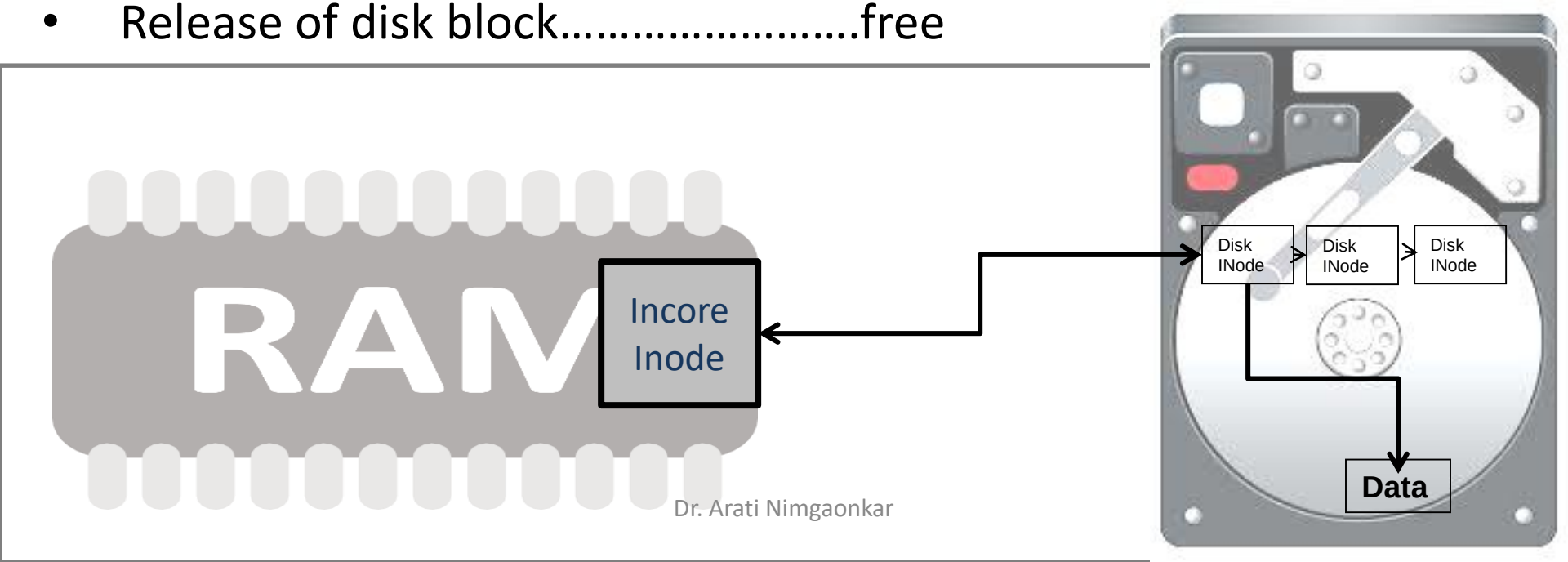
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- Allocating disk inode……ialloc
- Freeing disk inode……ifree



Algorithms related to inode

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- Release of in-core inode……iput
- Allocating disk inode……ialloc
- Freeing disk inode……ifree
- Allocation disk block……alloc
- Release of disk block……free



Algorithms related to inode

- Allocation of in-core inode……iget
- Release of in-core inode……iput
- Allocating disk inode……ialloc
- Freeing disk inode……..ifree
- Allocation disk block……alloc
- Release of disk block……free
- Converting filename to inode……namei